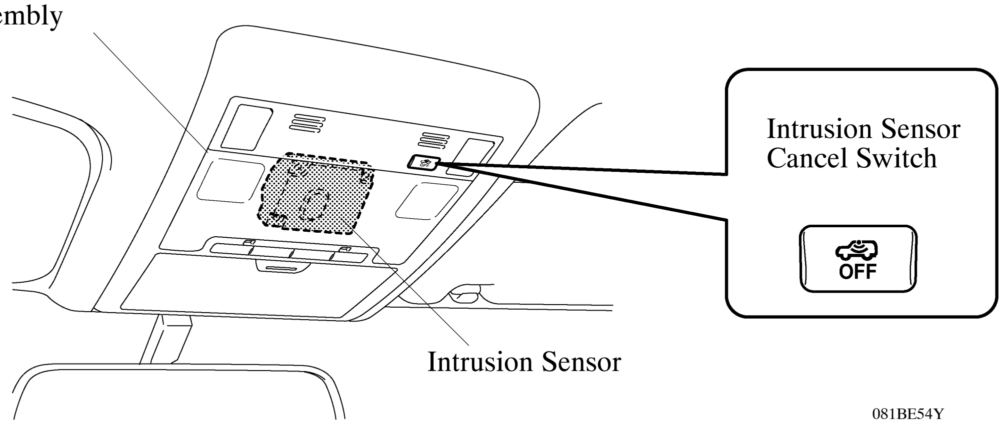
■ CONSTRUCTION AND OPERATION

1. Intrusion Sensor

General

- An intrusion sensor is provided in the front interior light located in the roof console box assembly.
- The intrusion sensor detects any intrusion into the vehicle interior and transmits a warning signal to the certification ECU.
- This sensor transmits radiowaves at 24.5 GHz in the vehicle interior. When an intruding object moves, it disrupts the reflection of the radiowaves, and the sensor detects the resulting changes in the phase of the radiowaves.



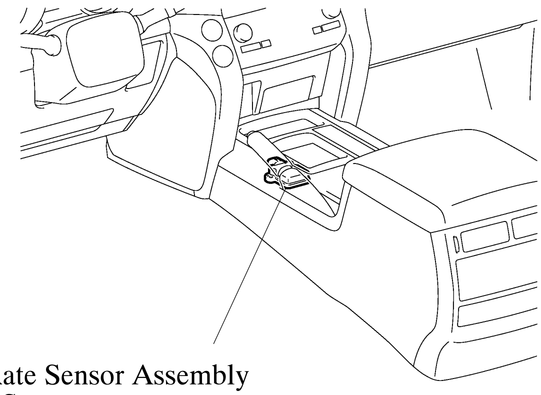
Operation

- The certification ECU activates the intrusion sensor when the system changes from the armed preparation state to the armed state. If an intrusion detection signal is received from the intrusion sensor during the armed state, the ECU changes to the alarm state in order to output an alarm.
- When the power source is OFF, pressing the intrusion sensor cancel switch can disable the function of the intrusion sensor. (For models with the tilt sensor, the tilt sensor function will also be disabled by pressing the cancel switch.) If the system changes to the armed state, the intrusion sensor will not activate. This function is canceled each time all doors are unlocked with a entry and start system, and is reinstated when the intrusion sensor is ON.
- When the key enters the actuation area of any door oscillator, the certification ECU judges and certifies the key ID code received from the tuner. At this time, while the certification ECU is evaluating that the key ID code matches the ID code stored in the memory, the intrusion sensor is disabled for approximately 30 seconds, thus preventing the sensor from detecting incorrectly and malfunctioning.

2. Tilt Sensor

General

- The tilt sensor is incorporated in the yaw rate sensor assembly located inside the front floor console.
- The tilt sensor detects pitch angles of vehicle inclination using the two deceleration sensors. When the tilt sensor detects a change in the pitch angle of vehicle inclination, such as attempting to tow the vehicle thieves, it transmits the warning signal to the certification ECU.



LHD Models

Operation

- The certification ECU activates the tilt sensor when the system changes from the armed preparation state to the armed state. If the warning signal is received from the tilt sensor during the armed state, the ECU changes to the alarm state in order to output an alarm.
- When the system is in the disarmed state, pressing the intrusion sensor cancel switch can disable the functions of the intrusion sensor and the tilt sensor simultaneously. If the system changes to the armed state, the tilt sensor will not activate. This function is canceled each time all doors are unlocked with a entry and start system, and is reinstated when the tilt sensor is ON.
- When the key enters the actuation area of any door oscillator, the certification ECU judges and certifies the key ID code received from the tuner. At this time, while the certification ECU is evaluating that the key ID code matches the ID code stored in the memory, the tilt sensor is disabled for approximately 30 seconds, thus preventing the sensor from detecting incorrectly and malfunctioning.

NOTE: Under the following conditions, if the vehicle tilts or an acceleration of gravity in the vertical and horizontal directions of the vehicle is detected by the tilt sensor, an alarm may occur accidentally. To avoid accidental alarm activation, turn the tilt sensor off using the tilt sensor OFF switch.

- The vehicle is transported by ferry, trailer or train.
- The vehicle is parked in multi-storied parking facilities.
- The vehicle is washed by a conveyor type car washing machine.
- A tire has blown out.
- The vehicle is jacked up for a tire or wheel change.
- An earthquake occurs or the road subsides.
- Goods such as ski boards and snowboards are loaded or unloaded.

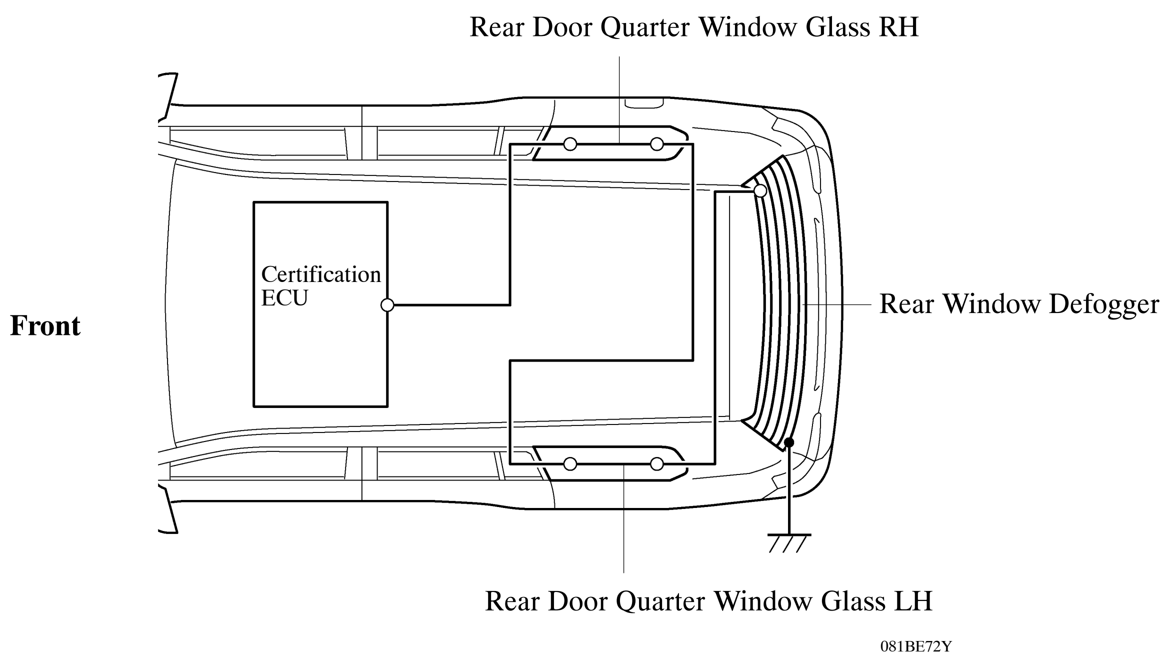
3. Self-power Siren

- A self-power siren is mounted on the rear quarter inner panel.
- During a normal operation, this siren sounds an alarm in accordance with the operation signal from the certification ECU. Because this siren has an internal battery, it can independently sound an alarm even if the vehicle's battery terminals are disconnected or if the wiring harnesses that are related to the system are cut.

4. Glass Print Pattern Type Glass Breakage Sensor

- The certification ECU, print patterns on the rear door quarter window glass and rear window defogger are connected in series, and when the certification ECU detects an open circuit, it determines that one of the windows has been broken.
- When the certification ECU receives an open circuit signal from the print pattern or rear window defogger for approximately 400 msec while in the armed state, it determines that one of the windows has been broken and changes the system to the alarm state in order to activate an alarm.
- When the key enters the actuation area of any door oscillator, the certification ECU judges and certifies the key ID code received from the tuner. At this time, while the certification ECU is evaluating that the key ID code matches the ID code stored in the memory, the print pattern and rear window defogger is disabled for approximately 30 seconds, thus preventing the sensor from detecting incorrectly and malfunctioning.

▶ System Diagram ◀



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